3.2 Basics of Polynomial Equations

3.2.1 Different types of Polynomial Equations

We already know that, for any non–negative integer n, a **polynomial** of **degree** n in one variable x is an expression given by

$$P \equiv P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \quad \ldots (1)$$

where $a_r \in \mathbb{C}$ are constants, $r = 0,1,2,\ldots,n$ with $a_n \neq 0$. The variable x is real or complex.

When all the coefficients of a polynomial P are real, we say **"P is a polynomial over $\mathbb{R}$"**. Similarly we use terminologies like **"P is a polynomial over $\mathbb{C}$"**, **"P is a polynomial over $\mathbb{Q}$"**, and **"P is a polynomial over $\mathbb{Z}$"**.

The function P defined by $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is called a **polynomial function.** The equation

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0 \quad \ldots (2)$$

is called a **polynomial equation**.

If $a_n c^n + a_{n-1}c^{n-1} + \cdots + a_1 c + a_0 = 0$ for some $c \in \mathbb{C}$, then c is called a **zero** of the polynomial (1) and **root** or **solution** of the polynomial equation (2).

If c is a root of an equation in one variable x, we write it as "$x = c$ is a root". The constants a_r are called **coefficients.** The coefficient a_n is called the **leading coefficient** and the term $a_n x^n$ is called the **leading term**. The coefficients may be any number, real or complex. The only restriction we made is that the leading coefficient a_n is nonzero. A polynomial with the leading coefficient 1 is called a **monic polynomial**.

Remark:

We note the following:

- Polynomial functions are defined for all values of x.
- Every nonzero constant is a polynomial of degree 0.
- The constant 0 is also a polynomial called the **zero polynomial**; its degree is not defined.
- The degree of a polynomial is a nonnegative integer.
- The zero polynomial is the only polynomial with leading coefficient 0.
- Polynomials of degree two are called **quadratic polynomials**.
- Polynomials of degree three are called **cubic polynomials**.
- Polynomial of degree four are called **quartic polynomials**.

It is customary to write polynomials in descending powers of x. That is, we write polynomials having the term of highest power (leading term) as the first term and the constant term as the last term.

For instance, $2x + 3y + 4z = 5$ and $6x^2 + 7x^2y^3 + 8z = 9$ are equations in three variables x, y, z; $x^2 - 4x + 5 = 0$ is an equation in one variable x. In the earlier classes we have solved trigonometric equations, system of linear equations, and some polynomial equations.

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